asthma

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Asthma Medications

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Introduction

Asthma is a wide-reaching, chronic inflammatory illness that impacts millions of people daily. It is frequently responsible for unscheduled healthcare usage, missed school, and workdays. It is an inappropriate immune response, much like an environmental allergy, to a triggering factor that induces bronchial hyperreactivity constriction with the remodeling of smooth muscle and increased mucous secretion into the airways.[1] Several classifications of medications are utilized to treat and manage chronic asthma to improve symptoms and reduce exacerbations. These include beta-2 agonists, anticholinergics, low-dose inhaled corticosteroids, medium-dose inhaled corticosteroids, high-dose inhaled corticosteroids, inhaled long-acting beta-agonists, leukotriene receptor antagonists, theophylline, cromolyn, zileuton, and newer class monoclonal antibody immune-modulating drugs. Oral corticosteroids may be used in an acute setting of asthma exacerbations.

Function

To better understand the role of various classes of medications in asthma it is very important to understand the basic physiology and pathophysiology of asthma. Asthma is a complex disease process that involves a multifaceted inappropriate immune reaction to an allergic antigen. It starts with an antigen-binding to dendritic cells. The cells process the antigen for presentation within the peripheral lymphoid tissue to naive helper T-lymphocytes (Th0). Th0 cells then trigger one of 2 immune cascades: a helper T-lymphocyte 1 (Th1) response or a helper T-lymphocyte 2 (Th2) response.

It appears that IL-12 is the determining factor for which the cascade pathway takes predominance. When IL-12 is present, Th1 cells are formed which leads to CD-8 cell-mediated immunity and neutrophilic mediated cytotoxic inflammatory responses that include the release of tissue necrosis factor and interferon gamma. When IL-12 is not present, a Th2 response occurs which leads to a more complicated cascade of interleukin and cytokine release from CD-4 cells including IL-4, IL-13, IL-9, IL-3, and IL-5.

Asthma is classically recognized as a typical Th2 disease, with increased IgE levels and eosinophilic inflammation in the airway. IL-4 and IL-13 trigger IgE production. IL-4 and IL-9 trigger mast cell activity. IL-3 triggers basophil activity. IL-5 is the primary trigger for eosinophil activity along with some input from IL-3 and GM-CSF. All components listed above are involved in the inflammatory response which leads to degranulation of histamine, prostaglandins, leukotrienes, among other enzymes. This induces bronchial hyper-responsiveness, increased mucous secretion, and ultimately airway obstruction and remodeling that is classic to asthma. Key points in this cascade that are currently susceptible to modulation include IgE and IL-5, while medications using IL-4 and IL-13 as targets are in clinical trials.

The initial binding of antigen-specific IgE to mast cell and basophil receptors sensitizes them for subsequent allergen exposure. Allergen re-exposure causes cross-linking of receptor-bound IgE, the immediate release of histamine and other granular-associated preformed mediators, as well as synthesis and release of lipid mediators, including leukotrienes and prostaglandins.

IL-5 is produced by Th2 cells and is necessary for the production, maturation, accumulation, and activation of eosinophils, as well as for their survival. In humans, IL-5 is primarily responsible for eosinophil proliferation and recruitment from bone marrow to ultimately aggregate in pulmonary tissue. Inhibiting IL-5 with monoclonal antibodies can reduce blood and bronchoalveolar eosinophilia caused by

allergic triggers or chronic asthmatic disease.

Once stimulated by IL-5, eosinophils recruit to the lung tissues and release proinflammatory interleukin and cytokine mediators and growth factors which ultimately lead to bronchial remodeling. This causes an overall upregulation of the inflammatory process. These chemical signals induce ultimate bronchial hyper-responsiveness and smooth muscle contraction, vascular leakage, hypersecretion of mucus, and shedding of epithelial cells. All of which are characteristics responsible for the ill effects of an asthma exacerbation. Various medications have already been developed and many, which work on different inflammatory pathways, are in development.[2][3][4][5]

Issues of Concern

Asthma is a very common illness in the developed world. It impacts approximately 23 million people in the United States with as many as 7 million of these being adolescents. Worldwide, over 300 million are impacted daily with as many as 250,000 deaths associated with asthma. In the United States, morbidity is greater in the Black as opposed to the White population.

Clinical Significance

The guidelines for determining appropriate therapy in the continued management of asthma are complicated and should be patient-centered following a stepwise progression of therapy intensity. These guidelines are reviewed in depth by the National Heart, Lung, and Blood Institute.First and foremost, the environmental factors should be considered, and appropriate avoidance techniques employed to reduce exposure to triggering substances.

Short and Long-Acting Inhaled Beta 2 Agonists

All patients diagnosed should be prescribed a short-acting, beta-2 agonist “rescue” inhaler. Most commonly, this is an albuterol metered-dose inhaler (MDI). Beta-agonist medications function by binding to beta-adrenergic receptors within the bronchioles. These receptors are coupled with stimulatory G proteins that activate adenylyl cyclase. Activation of adenylyl cyclase leads to an increase in intracellular cyclic adenosine monophosphate (cAMP). In smooth muscle tissue of the lung, cAMP decreases the total intracellular calcium content thus activating protein kinase A. In turn, this inactivates myosin light-chain kinase and activates myosin light chain phosphatase. This results in the relaxation of the smooth muscle. A side effect of this efflux of calcium is the activation of calcium-activated potassium channels in the cell membranes, which leads to hyperpolarization of the smooth muscle cells, inhibiting further activation and muscle contraction. The net effect of these actions is decreased contractility and decreased smooth muscle responsiveness to stimulus.

Inhaled, long-acting, beta-agonists most commonly used are salmeterol and formoterol. These medicines function by a similar mechanism of action to short-acting beta 2 agonists, except the half-life of the medicine is much longer, leading to a slower onset of action and a longer duration of effect. These should not be used as monotherapy due to an FDA black-box warning of the occurrences of severe asthma exacerbations, which in some patients with asthma is associated with death. Furthermore, monotherapy of a long-acting beta-agonist has not been seen to have sufficient benefit of therapy to justify the risk of adverse outcomes. This adverse risk is not seen in combination therapy with anticholinergic or corticosteroid medicines.[6][7]

Some side effects of beta-agonist medications have been noted to include tremor, increased nervousness and insomnia in children, nausea, fever, bronchospasm, vomiting, headache, pain, dizziness, cough, allergic reaction, dry mouth, sweating, chills, and dyspepsia.

Short and Long-Acting Inhaled Muscarinic Antagonists

Anticholinergic medicines, like beta-2 agonists, have 2 classes of either short-acting or long-acting duration of effect. Shorter-acting medicines typically used include ipratropium bromide, and longer-acting anticholinergic medicines include tiotropium, aclidinium, glycopyrronium, and umeclidinium. These medications function by binding to and blocking neural signals from parasympathetic muscarinic receptors M1, M2, and M3. M1 receptors are found on the cholinergic ganglia and function to modulate the neural transmission of a parasympathetic signal. M2 receptors are found on postganglionic nerve bulbs of parasympathetic nerve fibers. Blockage of this receptor functions to decrease acetylcholine release from the postganglionic nerve endings with a net decrease in signal transmission. M3 receptors are found on smooth muscle cells, mucosal glands, and vascular endothelium along the airways. Blockage of these receptors leads to a decrease in bronchoconstriction, mucus secretion from glandular tissues, and edema of the mucosal linings. While there is no specificity for which muscarinic receptor subtype is inhibited, the primary effect on asthma is found from inhibiting M3 receptors. Blocking of the receptor leads to a decrease in cyclic guanosine monophosphate (cGMP) content within the smooth muscle cells. In turn, there is a decrease in calcium content that activates protein kinase A inactivating myosin light-chain kinase and activating myosin light-chain phosphatase leading to smooth muscle relaxation. Decreases in intracellular calcium also function to decrease mucous secretion from glandular cells.[8][9]

Side Effects

Commonly encountered side effects from this group of medications are related to their systemic anticholinergic activity including urinary retention and lower urinary tract symptoms, particularly in elderly males, excessive dry mouth, headache, and dizziness. Others may include bronchitis, chronic obstructive pulmonary disease (COPD) exacerbation, sinusitis, dyspnea, urinary tract infection, headache, flu-like symptoms, back pain, cough, dyspepsia, and nausea.

Beta-2 agonist medications are used in combination with anticholinergics for a combined increased efficacy over monotherapy of either alone. Ipratropium bromide/albuterol is a common combination medication example of this strategy. Simultaneous administration of both an anticholinergic and a beta-2 agonist produces a greater bronchodilation effect than either drug is capable of achieving as monotherapy.[10]

Inhaled Corticosteroids

Inhaled corticosteroids are typically either once or twice daily inhaled medicines and include beclomethasone, budesonide nebules, flunisolide, fluticasone, and mometasone. These medications function to decrease the inflammatory response of an overactive immune system and effectively decrease the airway's hyperresponsiveness by inhibiting the production and release of chemotactic mediators and epithelial adhesion molecules necessary for the extravasation of immune cells into the airways. Furthermore, they function to decrease the survivability of inflammatory cells such as eosinophils, T lymphocytes, mast cells, and dendritic cells within the airways. Corticosteroids induce these effects on responsive cells by activating glucocorticoid receptors either directly or indirectly to regulate the transcription of specific target gene sequences in the nucleus of a cell. These modulations of transcription of DNA lead to suppression of inflammation by increasing the synthesis of anti-inflammatory proteins annexin-1, secretory leukoprotease inhibitor, and interleukin-10. Dosages can vary from minimal as a monotherapy to very high in combination therapy with either beta-2 agonists or anticholinergics. Common combination therapy medicines include fluticasone and salmeterol, budesonide and formoterol, mometasone and formoterol, and fluticasone and vilanterol. The goal of therapy is to achieve symptom relief with the smallest titration of a dose that is effective. As a result, dosing is based on clinical judgment of response, and if a patient has a good response, an attempt to reduce dosage to the lowest clinically efficacious dose should be made. Monotherapy with a corticosteroid is preferred to combination therapy with a beta-2 agonist because corticosteroids offer control of the underlying inflammatory pathology, thus reducing recurrence of exacerbations; whereas, beta-agonists provide symptomatic relief of asthma. If control is possible with monotherapy, this gives better long-term outcomes, has fewer potential side effects, and reduced the cost of medication long term.[11]

Side Effects

Side effects that may occur with the use of inhaled corticosteroids include nasopharyngitis, headache, bronchitis, sinusitis, influenza, pharyngitis, respiratory tract infection, oropharyngeal pain, toothache, back pain, viral gastroenteritis, abdominal pain, cough, oropharyngeal candidiasis, dysphonia, oral candidiasis, procedural pain, rhinitis, and throat irritation.

Leukotriene Receptor Antagonists

Leukotriene receptor antagonists include montelukast and zafirlukast. These medications work to antagonize the effects of pro-inflammatory leukotrienes in the inflammatory response of the immune system, functionally decreasing the ability of mast cells to degranulate releasing pro-inflammatory factors and chemotactic signals for a further immune response. This results in decreased inflammation and decreased responsiveness of the airways to immune triggers.

Side Effects

Some side effects that may occur include headache, abdominal pain, eczema, influenza, laryngitis, pharyngitis, viral infection, wheezing, dental pain, dizziness, dyspepsia, elevated liver function tests, fever, gastroenteritis, nasal congestion, otitis, rash, urticaria, bronchitis, cough, sinusitis, upper respiratory tract infection, allergic granulomatous angiitis (Churg-Strauss syndrome), cholestatic hepatitis, and rarely, aggressive behavior or suicidal thoughts.

Methylxanthine

Theophylline is a methylxanthine drug that functions through two actions. It decreases the production and release of pro-inflammatory signals such as TNF-alpha and leukotriene, effectively decreasing inflammation, and it acts as a direct adenosine receptor antagonist nonspecifically leading to smooth muscle relaxation and dilation of the bronchioles reducing airway obstruction.[12]

Side Effects

Although the frequency has not been defined, some possible side effects for theophylline include central nervous system excitement, headache, insomnia, irritability, restlessness, seizure, diarrhea, nausea, vomiting, diuresis, exfoliative dermatitis, skeletal muscle tremors, tachycardia, cardiac flutter, hypercalcemia, difficulty urinating, acute myocardial infarction, seizures, and urinary retention. Furthermore, the therapeutic level in serum is narrow with a narrow window between the therapeutic and toxic range. It, therefore, requires close monitoring of serum levels with periodic blood work.

Cromolyn and Zileuton

Cromolyn is a mast cell stabilizing medication that functions to decrease the degranulation of proinflammatory cytokines such as histamine. Furthermore, it is thought to decrease the neural response to irritation of sensory nerve fibers in the airways and decrease the release of cytokines from pre-formed eosinophils.

Zileuton is a 5-lipoxygenase inhibitor medication that functions to decrease leukotriene production, effectively decreasing the inflammatory response of the immune system.

Side Effects

Possible side effects for cromolyn include cough, flushing, palpitation, chest pain, nasal congestion, nausea, fatigue, migraine, sneezing, wheezing, psychosis, pruritus, dysphagia, esophagus spasm, pancytopenia, polycythemia, tinnitus, and pharyngitis.

Monoclonal Antibody Immune-Modulating Drugs

Newer class monoclonal antibody immune-modulating drugs can be used in specific situations where patients do not respond effectively to inhaled glucocorticoid therapy and are found to have eosinophilia. These medications aim to decouple the Th2 inflammatory pathway to decrease the immune response to a triggering event, effectively decreasing eosinophilia.

Omalizumab may be an option for adults and adolescents 12 years of age and older who have moderate to severe asthma, a positive skin or in vitro test result in response to a perennial aeroallergen, and symptoms not adequately controlled by inhaled corticosteroids. The dose and frequency of administration (every 2 or 4 weeks) of omalizumab are based on the initial total IgE levels (greater than 30 to 700 IU/ml) and body weight. This medicine functions to directly bind free IgE, thus reducing its ability to signal further Th2 response including eosinophilia and mast cell activation. Baseline IgE levels are not predictive of response but are needed for patient selection and dose determination.

Side Effects

Possible side effects of omalizumab include injection site reactions, viral infections, upper respiratory infection, sinusitis, headache, pharyngitis, pain, arthralgia, fracture, fatigue, dermatitis, arm pain, leg pain, dizziness, earache, pruritus, nasopharyngitis, pyrexia, upper abdominal pain, streptococcal pharyngitis, otitis media, viral gastroenteritis, epistaxis, alopecia, edema, anaphylaxis, bronchitis, and urticaria.

Mepolizumab is an add-on treatment indicated for patients 12 years of age and older with severe asthma and high eosinophil levels at 150 cells/mcL or greater pre-treatment. Dosage is 100 mg subcutaneously every four weeks. As discussed above, this medicine binds to IL-5, thus inhibiting its signal to proliferate eosinophils in the bone marrow.[13]

Side Effects

Some possible side effects of mepolizumab include headache, injection site reactions, systemic allergic/nonallergic reactions, back pain, fatigue, systemic allergic/hypersensitivity reactions, influenza, urinary tract infection, upper abdominal pain, pruritus, eczema, and muscle spasms.

Reslizumab is an add-on therapy for adults ages 18 years and older with severe asthma and eosinophilia. Dosage is 3 mg/kg intravenously infused every four weeks. As discussed above, this medicine binds to IL-5, thus inhibiting its signal to proliferate eosinophils in the bone marrow.

Side Effects

Possible side effects of reslizumab include elevated creatine phosphokinase (CPK), oropharyngeal pain, myalgias, and anaphylaxis.

Benralizumab 100 mg given subcutaneously every eight weeks is currently in human trials, but data looks promising for treatment in refractory asthma and COPD. Similar to mepolizumab and reslizumab, this medicine binds to the IL-5 receptor, thus inhibiting its signal to proliferate eosinophils in the bone marrow and augmenting eosinophil apoptosis. The extent of reduction in exacerbations, improvement in FEV1, and symptom reduction seem proportional to the baseline blood eosinophil count.

Side Effects

Possible side effects of benralizumab include headache, pharyngitis, pyrexia, and various hypersensitivity reactions.

Lebrikizumab, a monoclonal antibody against IL-13 failed to provide consistent benefit in asthmatics with type-2 inflammation. Tralokinumab, another potential therapeutic agent with the identical target, is in a clinical trial (NCT02194699 and NCT02281357).

Dupilumab, a human monoclonal antibody to the IL-4 receptor, blocks both IL-4 and IL-13 signaling. Wenzel et al. demonstrated a 60% to 80% reduction in exacerbations in asthma patients already on inhaled corticosteroid-LABA combination. The responders did have higher baseline IgE levels. It is yet to be FDA approved for asthma.

Bronchial Thermoplasty

In patients with severe persistent asthma that is poorly or non-responsive to medical therapy, another non-medical approach is possible called bronchial thermoplasty. This is the delivery of controlled, therapeutic radiofrequency energy to the airway wall. It heats and destroys bronchial tissue. The epithelium, blood vessels, mucosa, and nerves of the airway will recover and regenerate over time. However, airway smooth muscle does not regenerate and is replaced with connective tissue. The net result is a decrease in bronchoconstriction due to loss of musculature.

Patient selection and appropriate time of delivery is the key as it can cause a post-procedure exacerbation of bronchospasm and should be avoided in individuals with ongoing exacerbation or with chronic changes of interstitial disease and bronchiectasis. The clinical trials excluded patients with 3 or more exacerbations per year, FEV1 below 60%, and chronic rhinosinusitis.[5][14]

Enhancing Healthcare Team Outcomes

Asthma is a very common disorder that affects people of all ages. Even the most well-controlled asthma patient can rapidly develop a serious attack that can be life-threatening. The condition is best managed by an interprofessional team that includes a pulmonologist, internist, pharmacist, nurse practitioner, emergency room personnel, and an intensivist. The key to decreasing patient morbidity is education. At every opportunity, the patient must be told to control the environmental trigger factors and remain compliant with the prescribed medications. The pharmacist should regularly ask the patient about symptoms as it may reflect poor drug compliance or worsening of the disease. The pharmacist should always check for potential drug-drug interactions or medications accidentally prescribed that can exacerbate asthma and communicate concerns to the interprofessional team managing the patient. Similarly, the nurse should carefully examine all asthmatic patients for wheezes and ensure that they follow up with the pulmonologist regularly. There are a vast number of asthma medications on the market, and thus, the pharmacist should keep a track of what the patient is taking and ensure that no adverse reactions are developing. Today, many communities have an asthma clinic staffed by a nurse, where the disorder is monitored, and the patient is educated. Finally, all members of the team should coordinate the education of asthmatics as they should be encouraged to take their daily medications, discontinue smoking, and refrain from keeping pets. (Level 1)[15][16]

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Level 1 (high-level) evidence

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